

Sayantani Bhattacharya

Department of Mathematics and Statistics, Auburn University, AL, 36849

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Education

Auburn University, Alabama, USA
PhD in Mathematics

Aug 2023 – Present

Tezpur University, Assam, India
Master's in Mathematics

2018 – 2020

Chennai Mathematical Institute, Chennai, India
Bachelor's in Mathematics and Computer Science (Dual Degree)

2015 – 2018

Research Interests

Matroid Theory, Combinatorics.

Preprints & Publications

1. S. Bhattacharya, J. D. Clifton, Z. Walsh, The unbreakable quasi-graphic matroids. **Submitted.** [\[arxiv\]](#)

Talks

- ◇ Combinatorics Seminar, Auburn University, Auburn, AL (8th April, 2026)
The Unbreakable Quasi-graphic Matroids
- ◇ 34th Cumberland Conference on Combinatorics, Graph Theory and Computing, Auburn University, Auburn, AL (17th May, 2026)
The Unbreakable Quasi-graphic Matroids

Teaching Experience

Graduate Teaching Assistant, Auburn University, AL

<i>MATH 2630 (Section: R10, R11) (Calculus III) : Recitation Leader.</i>	Fall 2026
<i>MATH 1610 (Section: R25, R26) (Calculus II) : Recitation Leader.</i>	Spring 2026
<i>MATH 1950 MR1 (Merit Recitation Program, Calculus I) : Instructor of Record.</i>	Fall 2025
<i>MATH 1610 (Section: R15, R16) (Calculus I) : Recitation Leader.</i>	Spring 2025
<i>MATH 1000 (Section: 005, 006) (College Algebra) : Instructor of Record.</i>	Fall 2024
<i>MATH 1000/1120/1130/1150/2650 : GTA (Tutoring Center).</i>	Summer 2024
<i>MATH 1610 (Calculus I) : Grader.</i>	Spring 2024
<i>MATH 2650 (Linear Differential Equations) : Grader.</i>	Fall 2023

Graduate Research Assistant, Auburn University, AL

<i>Research Assistant (funded under Dr. Zachary Walsh)</i>	Summer 2026
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Fellowships & Awards

- ◇ Awarded \$1000 towards Professional Development for the completion of the Auburn University Summer STEM Institute (AUSSI 2025) Training Program.
- ◇ Graduate Teaching Assistantship and Fellowship, Auburn University (2023–Present)
- ◇ Council of Scientific and Industrial Research (CSIR India) National Eligibility Test (NET) with Junior Research Fellowship (JRF) Awardee (Nov 2019)
- ◇ INSPIRE Scholarship for Higher Education (SHE) Fellowship, Govt. of India (2015–2018)
- ◇ Institutional Fellowship at Chennai Mathematical Institute for Undergraduates (2015–2018)

Workshops & Field Schools

- ◇ **Professional Development for Emerging Education Researchers (PEER) Institute Field School**, Auburn University, June 2026
- ◇ **Science Communication Workshop Series**, Biggio Center, Auburn University, Fall 2025
- ◇ **Topological Methods in Combinatorics Summer School**, Iowa State University, May 2025
- ◇ **Advanced Instructional School on Advanced Linear Algebra (AIS-ALA 2022)**, ATM Schools, National Center for Mathematics, Mumbai, India, May, 2022

Service & Outreach

- ◇ **Association for Women in Mathematics (AWM) at Auburn University**
 - President, 2026–27
 - Treasurer, 2025–26
- ◇ Secretary, **Mathematical Sciences Graduate Student Council (MSGSC)**, Auburn University, 2026–27
- ◇ DMS Volunteer, **34th Cumberland Conference**, Auburn University (May 15–17th, 2026)
- ◇ **Mathematical Puzzle Programs (MaPP)**, Auburn University
 - Volunteer, Feb 21st, 2026
 - Volunteer, Feb 15th, 2025
- ◇ **Alabama State Science Olympiad**, Auburn University
 - Volunteer, Bungee Drop Div C, Feb 14th, 2026
 - Volunteer, Codebusters Div B, Mar 29th, 2025
- ◇ **Destination STEM Expo**, Auburn University
 - Exhibitor (Chess game : Rook's Move; Puzzle game : Gerrymandering), Oct 1st, 2026
 - Exhibitor (Model : Brachistochrone Curve), Oct 15th, 2025

- ◇ Volunteer, **Julia Robinson Mathematics Festival (JRMF) at Joint Mathematics Meeting (JMM)**, Washington DC (Jan 4th, 2026)
- ◇ Mentor, **First Year Experience (FYE) Mentorship Program**, Graduate Student Council, Auburn University (since Feb 2025)
 - ▷ *Awarded the Outstanding First Year Experience (FYE) Mentor Award 2025 by the Auburn University Graduate School.*
- ◇ President, **College of Science & Mathematics International Students Association (CISA)**, Auburn University (Nov 2023 – Aug 2024)

Technical Skills

Programming: Python, C, SageMath, LaTeX